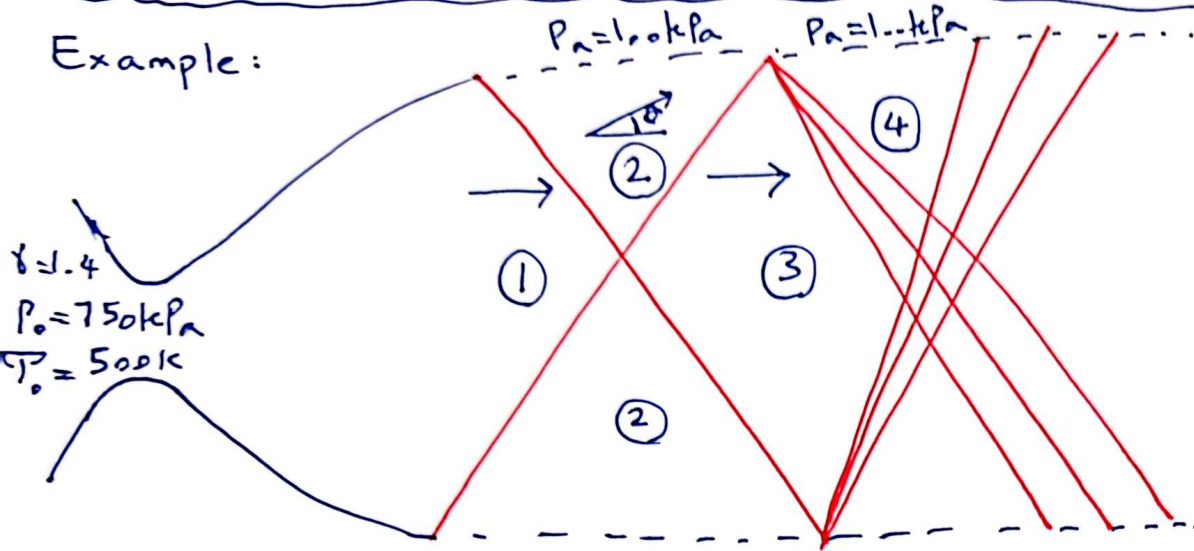


WED- Feb-19-2025: Example of over-expanded Nozzle

Example:



$$A_e = 0.002637 \text{ m}^2$$

$$A_t = 0.001 \text{ m}^2$$

Find flow conditions (M, T, P) in (1) (2) (3) (4)

Solution: $\frac{A_e}{A^*} = 2.637 \xrightarrow{\text{online calculator}} M_1 = 2.5$

$$P_1 = \left(\frac{P_1}{P_0} \right) \times P_0 = 0.0585 \times 750 \text{ kPa} = 43.875 \text{ kPa}$$

online calculator $T_1 = \left(\frac{T_1}{T_0} \right) \times T_0 = 0.444 \times 500 \text{ K} = 222.2 \text{ K}$

Section 2 is confined by ambient

$$So \quad P_2 = P_a = 100 \text{ kPa}$$

(1) (2)
 $M_1 = 2.5$
 $P_1 = 43.875 \text{ kPa}$
 N-S table

$$\frac{P_2}{P_1} = \frac{100}{43.875} = 2.2792$$

$$\beta = 35.39^\circ \rightarrow \text{Wave angle}$$

M_1	$\frac{P_2}{P_1}$
M_{n1}	2.2792

if using table $\left\{ \begin{array}{l} M_{n1} = M_1 \sin \beta \\ \sin \beta = \frac{M_{n1}}{M_1} = \frac{\boxed{}}{2.5} \end{array} \right.$

(P-1)

$$\begin{cases} \beta = 35.39^\circ \\ M_1 = 2.5 \end{cases} \quad \text{online cal. for shock} \quad \theta = 13.55 \text{ (turn angle)} \quad M_2 = 1.9362 \quad P_2 = 1.0 \text{ kPa}$$

$$\frac{T_2}{T_1} = 1.2858$$

$$T_2 = \frac{T_2}{T_1} \times T_1 = 1.2858 \times 222.2 = \underline{285.704 \text{ K}}$$

~~$P_2 = 1.0 \text{ kPa}$ ②
 $T_2 = 285.7 \text{ K}$
 $M_2 = 1.9362$~~ ③

The key point about reflected oblique shock ③ is that it again turns the flow to the horizontal direction.

online calc. $\rightarrow$

$$\begin{cases} M_2 = 1.9362 \\ |\theta_3 - \theta_2| = |0 - 13.55| = 13.55 \\ \beta \text{ (wave angle)} = 44.98^\circ \end{cases}$$

$M_3 = 1.44$

$$\frac{P_3}{P_2} = 2.018 \rightarrow P_3 = 2.018 \times 1.0 = \underline{2.018 \text{ kPa}}$$

$$\frac{T_3}{T_2} = 1.23 \rightarrow T_3 = \frac{T_3}{T_2} \times T_2 = 1.23 \times 285.7 = \underline{352.7 \text{ K}}$$

$$T_0 = T + \frac{u^2}{2c_p}$$

decreases

(P-2)

③ ④ $P_4 = 100 \text{ kPa}$

$M_3 = 1.438$

$P_3 = 201.86 \text{ kPa}$

$T_3 = 352.7 \text{ K}$

strategy: compute $\frac{P_3}{P_4} = 2.0186$

$P_{03} = P_{04}$, find M_4 from:
isentropic

$$\frac{P_3}{P_4} = \frac{P_3/P_{03}}{P_4/P_{04}} = \frac{(1 + \frac{\gamma-1}{2} M_3^2)^{-\frac{\gamma}{\gamma-1}}}{(1 + \frac{\gamma-1}{2} M_4^2)^{-\frac{\gamma}{\gamma-1}}} \rightarrow \text{unknown}$$

$$= \left(\frac{1 + \frac{\gamma-1}{2} M_4^2}{1 + \frac{\gamma-1}{2} M_3^2} \right)^{\frac{\gamma}{\gamma-1}}$$

known

Solve the above relation for M_4 ,

$M_4 = 1.907$

$P_4 = 100 \text{ kPa}$

$T_4 = \left(\frac{T_4}{T_{04}} \right) \times T_{01} = 0.578 \times 500 = 288.6 \text{ K}$

$M_3 = 1.438 \rightarrow \psi(M_3) = 10.09$

$M_4 = 1.907 \rightarrow \psi(M_4) = 23.78 \rightarrow \Delta\psi = \psi(M_4) - \psi(M_3) = 13.7^\circ$